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Beata Zalewa

Dos and Don'ts for Azure Serverless with Azure Functions and Logic Apps

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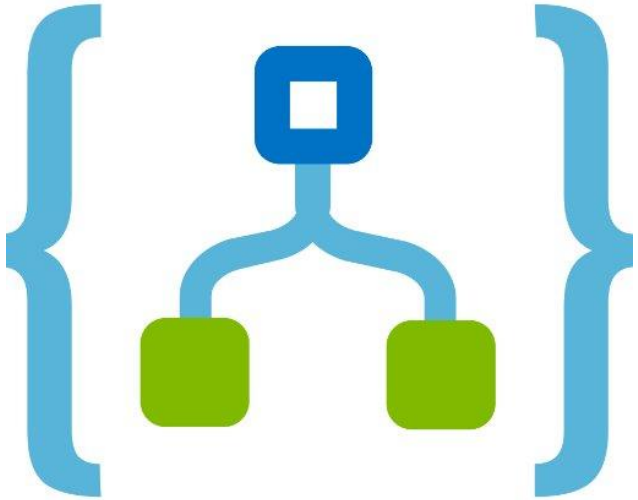

Dos and Don'ts for Azure Serverless with Azure Functions and Logic Apps

Beata Zalewa, Expert Summit 2020, 05.11.2020

ABOUT

/bzalewa

In professional life consultant, Microsoft Certified Trainer, administrator, developer, freelancer, sometimes a miracle-worker and magician.

From the beginning of career associated with Microsoft technologies.

Private life in numbers: 1 husband, 1 daughter, 1 cat and 2 dogs. My hobbies are detective stories and photography.

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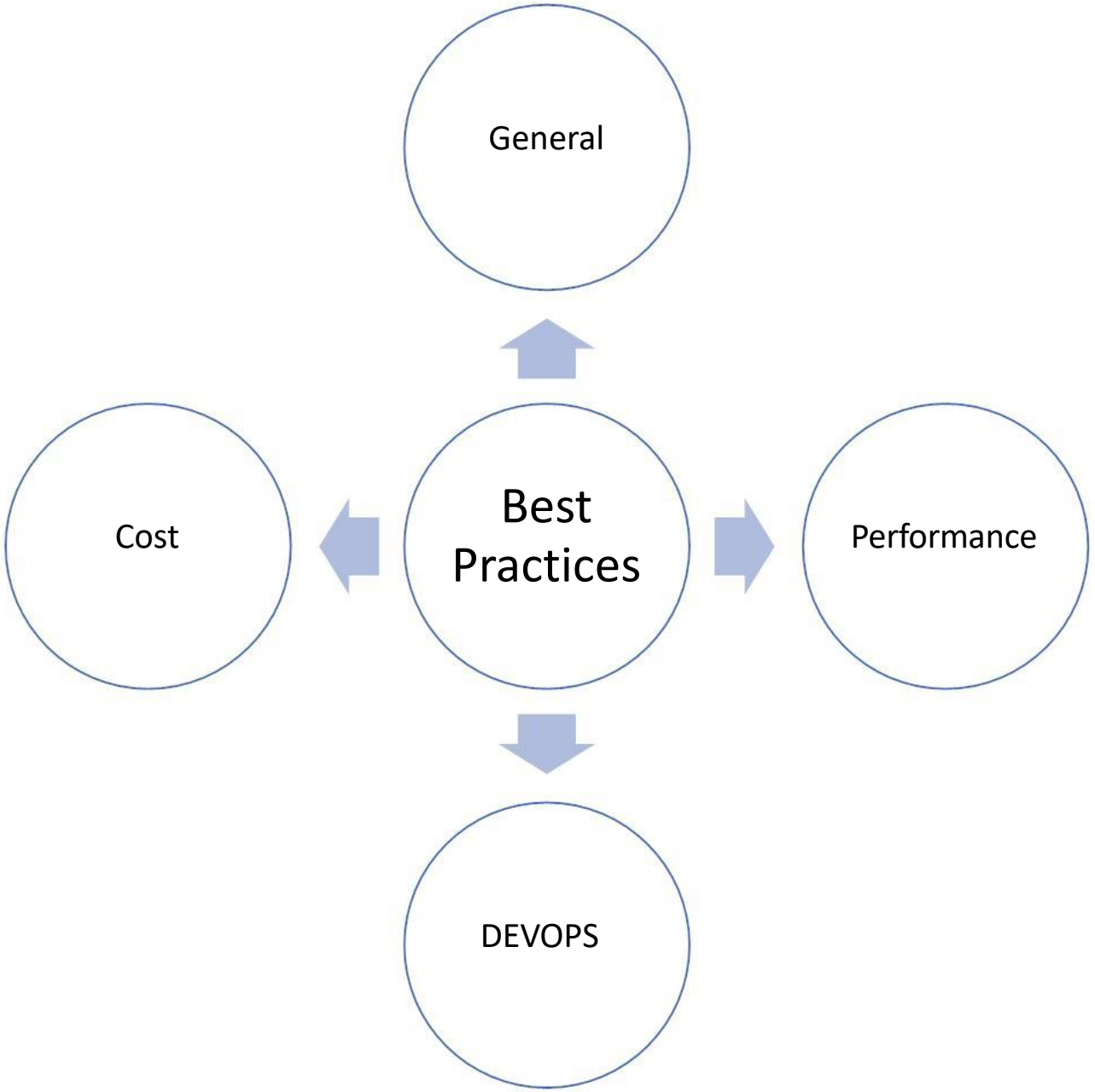
General
Cost
Best Practices
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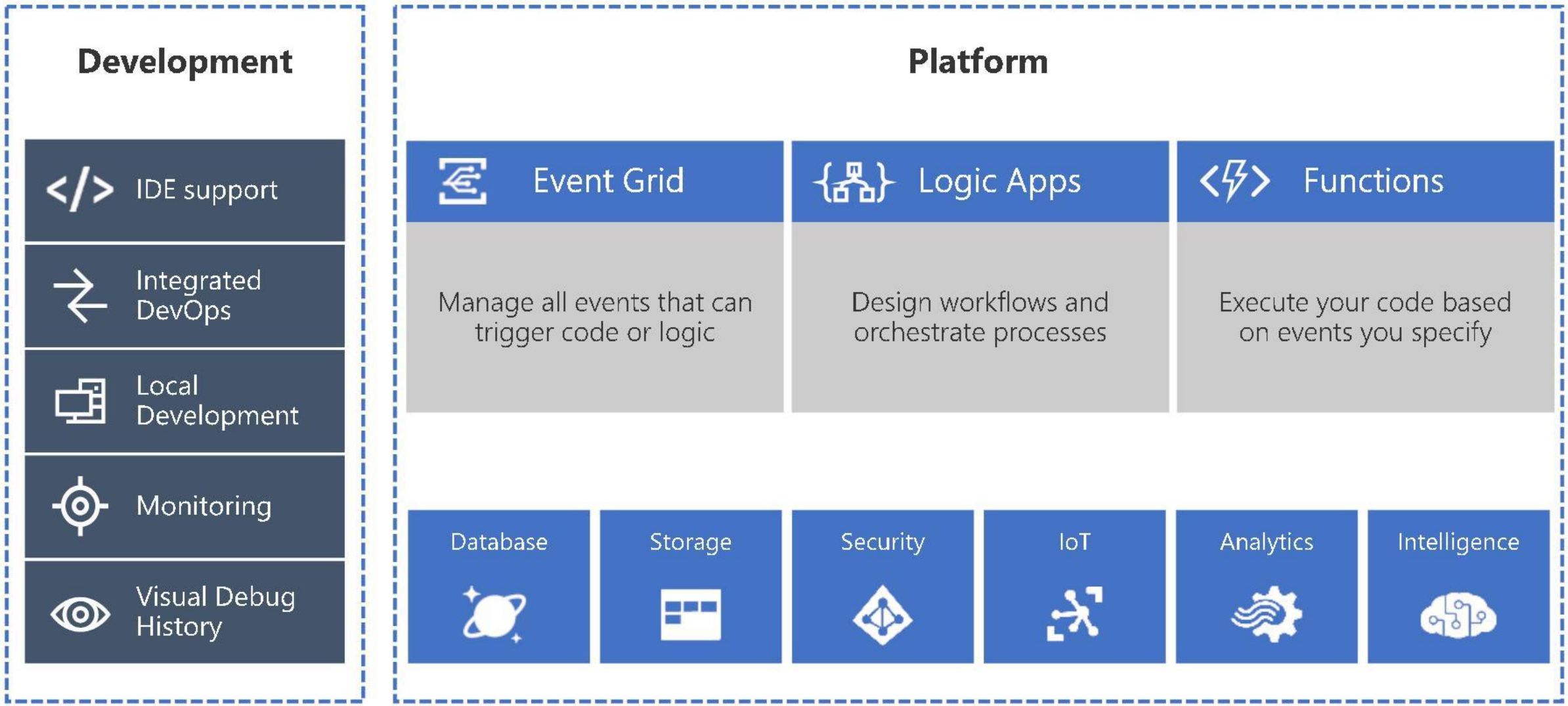
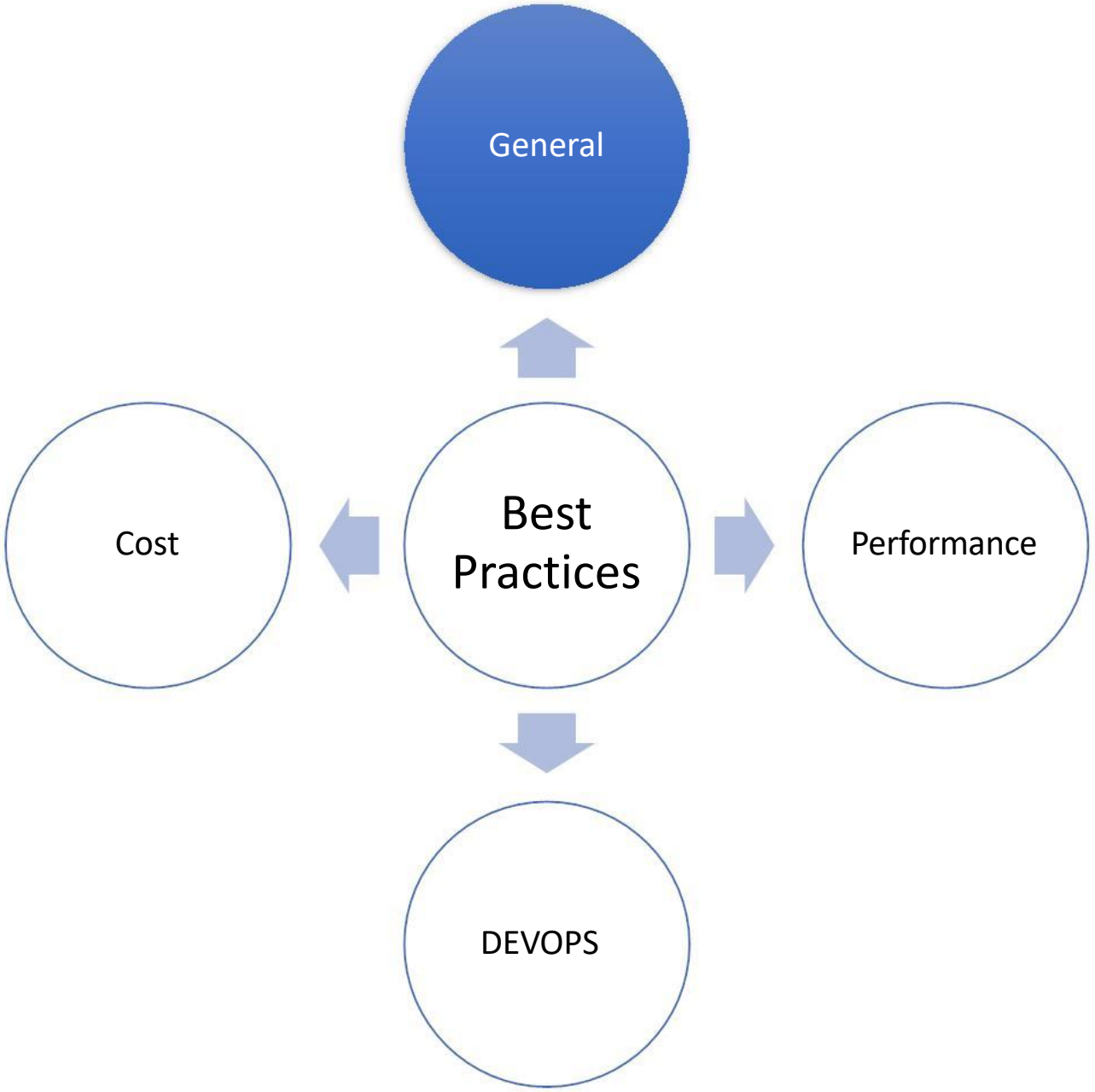
Serverless platform components




General
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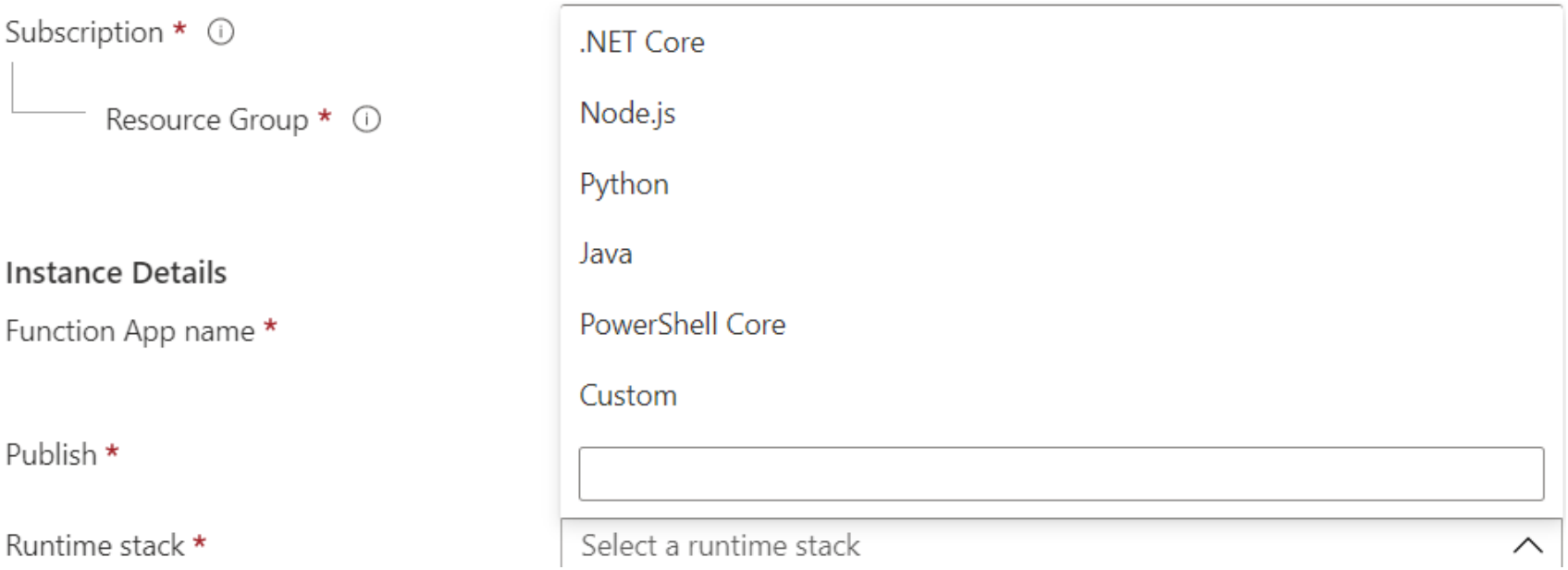
Azure Functions

Solution for running small pieces of code, or "functions," in the cloud:

- Write only code that is relevant to business logic
- Removes the necessity to write “plumbing” code to connect or host application components

Build on open-source WebJobs code

Supports a wide variety of programming languages, for instance, even supports scripting languages, such as:



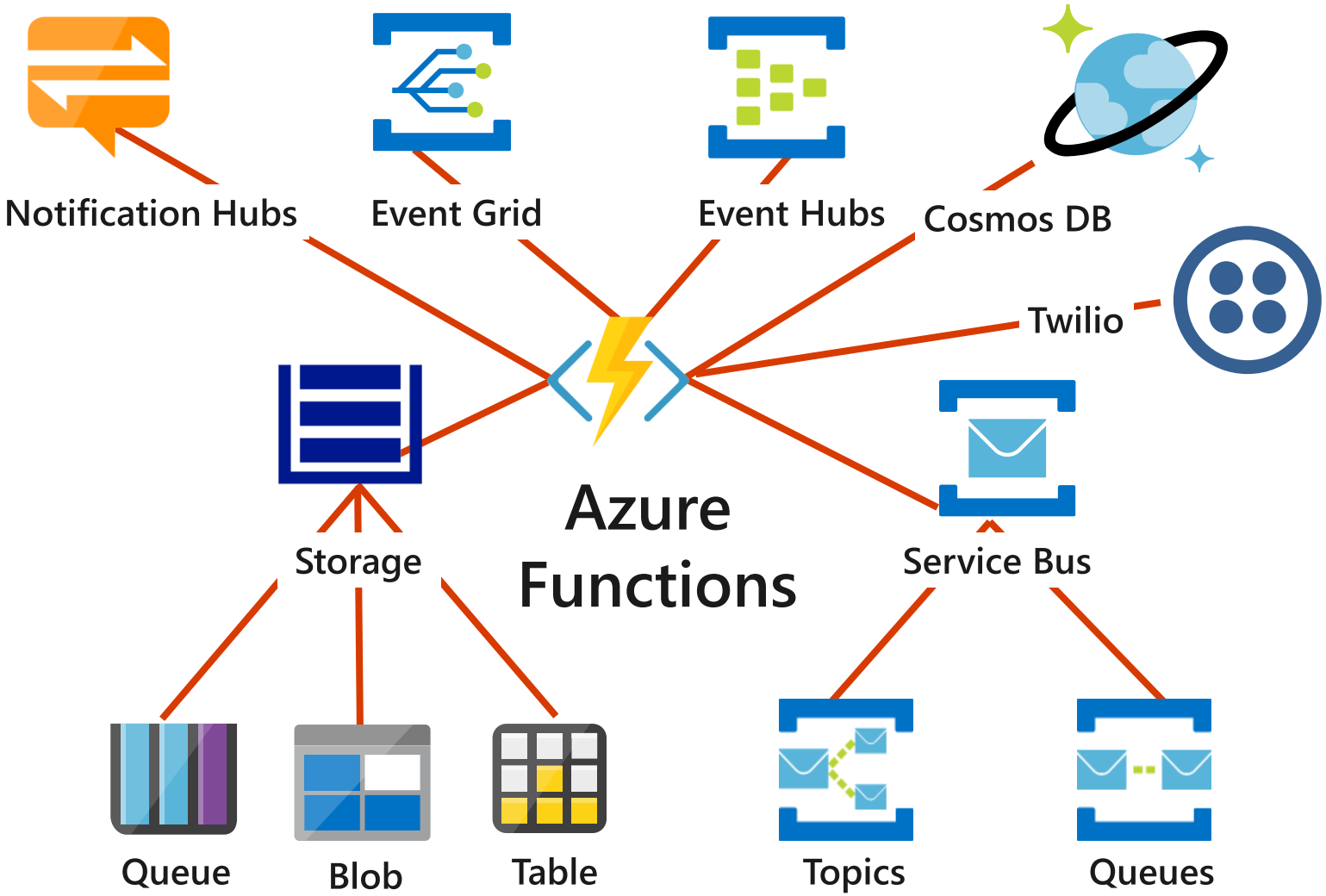
What can Azure Functions do?

- Run code based on HTTP requests
- Schedule code to run at predefined times
- Process new and modified:
  - Azure Cosmos DB documents
  - Azure Storage blobs
  - Azure Queue storage messages
- Respond to Azure Event Grid events by using subscriptions and filters
- Respond to high volumes of Azure Event Hubs events
- Respond to Azure Service Bus queue and topic messages

Trigger types

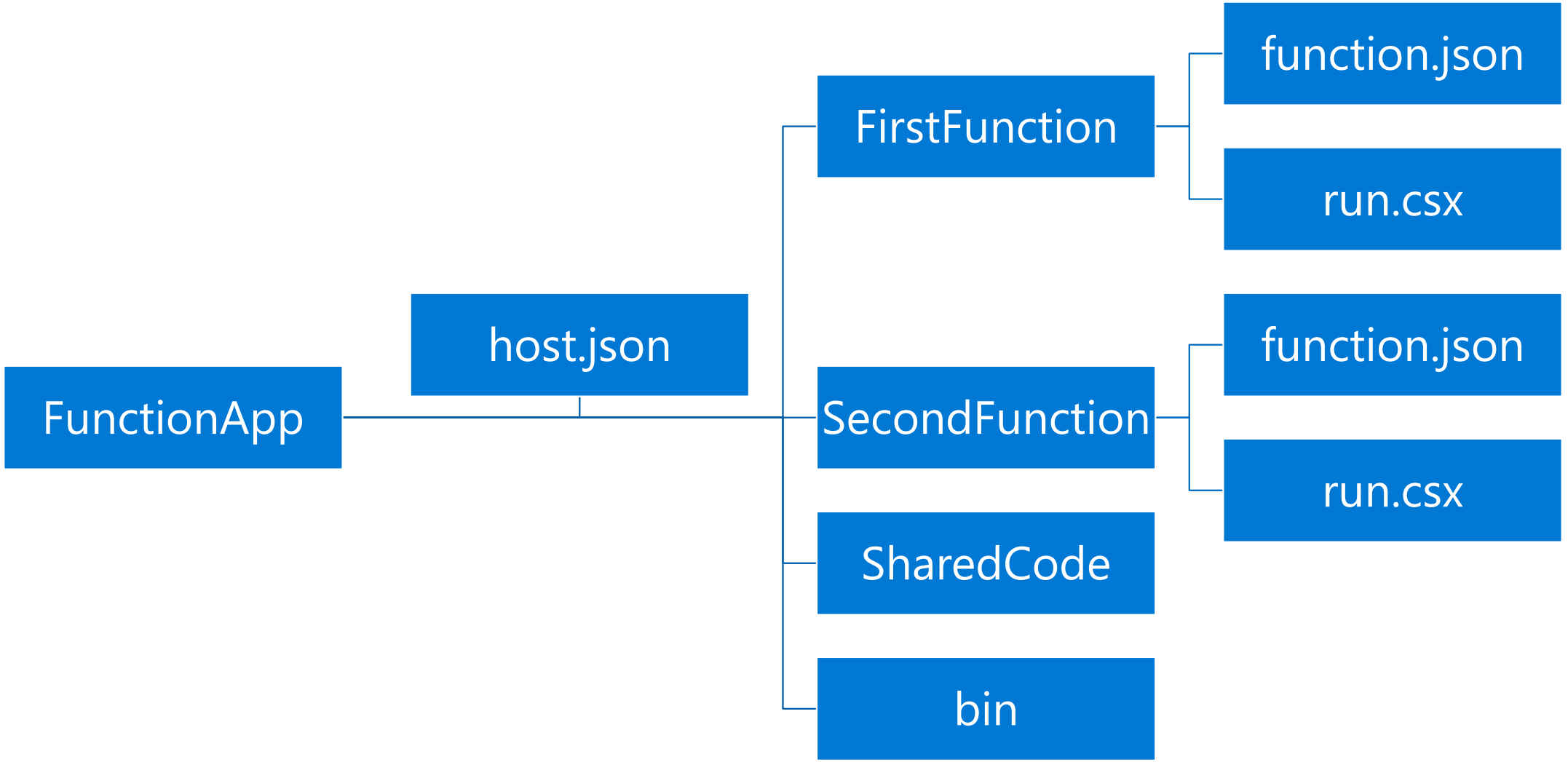
- Triggers based on Azure services:
  - Cosmos DB
  - Blob and queues
  - Service Bus
  - Event Hub
- Triggers based on common scenarios:
  - HTTP request
  - Scheduled timer
- Triggers based on third-party services:
  - GitHub
- And more…

Function integrations



DEMO 1

Function folder structure

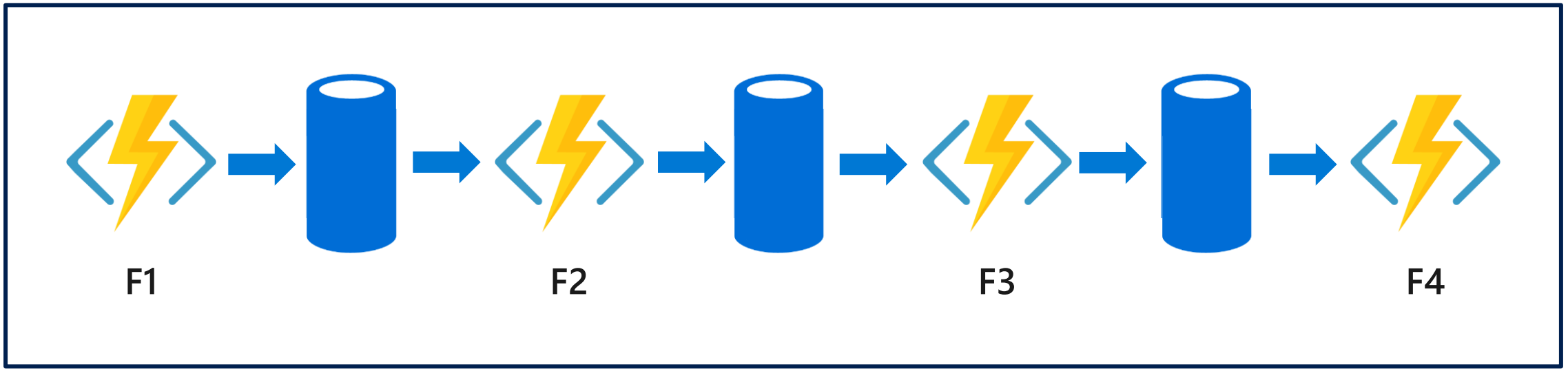


Durable Functions

- Write stateful functions in a stateless environment
- Manages state, checkpoints, and restarts
- Defines an Orchestrator function
  - Workflows are defined in code
  - Calls other functions synchronously or asynchronously
  - Checkpoint progress whenever function awaits

Durable Function scenario - Chaining

Function chaining refers executing a sequence of functions in a particular order. Often, the output of one function needs to be applied to the input of another function.



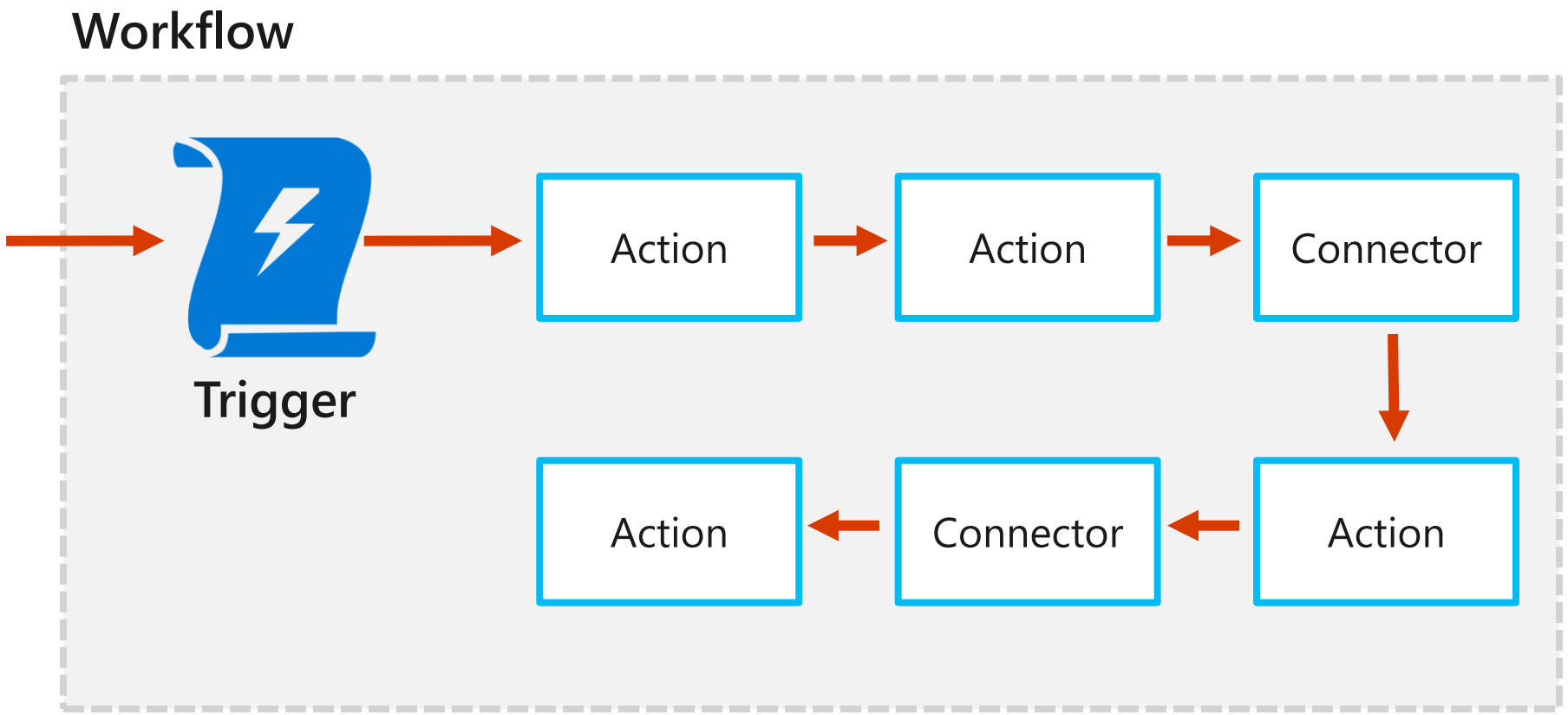
Azure Logic Apps

- Automation workflow solution:
  - No-code designer for rapid creation of integration solutions
  - Prebuilt templates to simplify getting started
  - Out-of-box support for popular software as a service (SaaS) and on-premises integrations
  - BizTalk APIs available to advanced integration solutions
- JSON-based workflow definition:
  - Can be deployed by using Azure Resource Manager templates

Components

- Workflow
  - The business process described as a series of steps
- Triggers
  - The step that invokes a new workflow instance
- Actions
  - An individual step in a workflow, typically a Connector or custom API app
- Connectors
  - A special case of an API app that is prebuilt and ready to integrate with a specific service or data source. For example:
    - Twitter and SQL Server Connectors

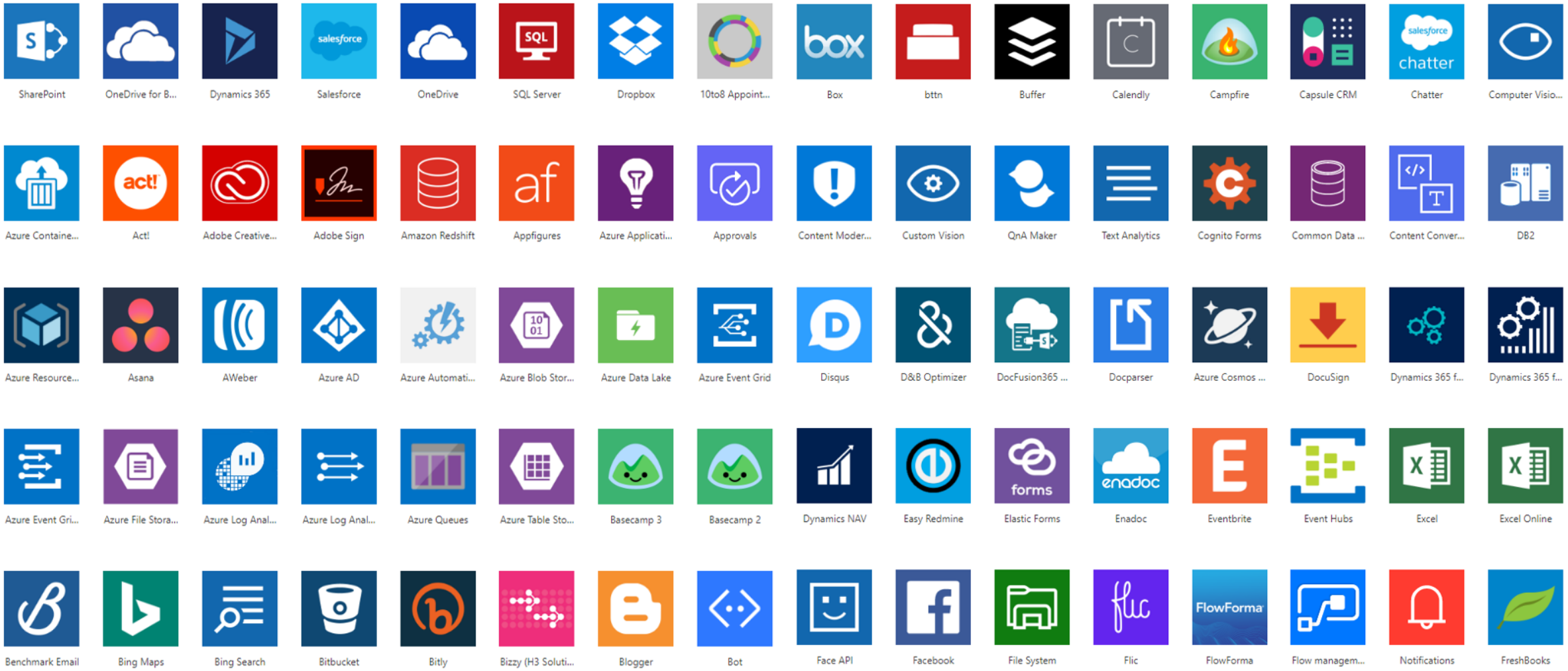
Workflow components



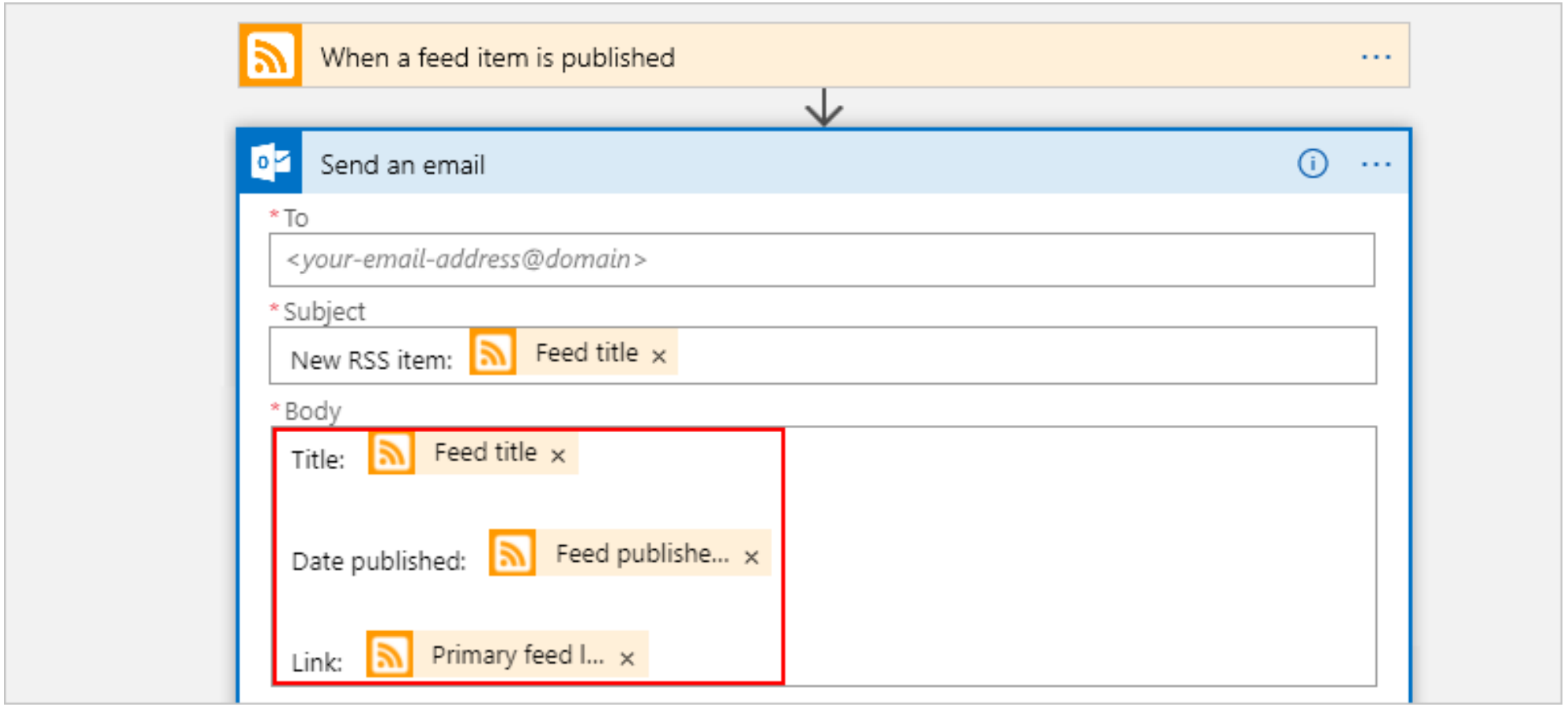
Connectors


Facebook Connector
Microsoft
Dropbox Connector
Microsoft
Office365 Connector
Microsoft
Salesforce Connector
Microsoft
SharePoint Server Connector
Microsoft
Twitter Connector
Microsoft
FTP Connector
Microsoft
HTTP Listener
Microsoft
Microsoft SQL Connector
Microsoft
MQ Connector
Microsoft
Informix Connector
Microsoft
Oracle Database Connector
Microsoft
BizTalk Flat File Encoder
Microsoft
BizTalk JSON Encoder
Microsoft
BizTalk XML Validator
Microsoft
BizTalk Transform Service
Microsoft
BizTalk X12
Microsoft
BizTalk XPath Extractor
Microsoft
Template deployment
Microsoft
AS2 Connector
Microsoft
Azure Service Bus Connector
Microsoft
Azure WebJobs
Microsoft
BizTalk EDIFACT
Microsoft
Azure Storage Blob Connector
Microsoft
SMTP Connector
Microsoft
Wait
Microsoft
Twilio Connector
Microsoft
SugarCRM Connector
Microsoft
Slack Connector
Microsoft
Yammer Connector
Microsoft
Box Connector
Microsoft
BizTalk Rules
Microsoft
Chatter Connector
Microsoft
DB2 Connector
Microsoft
File Connector
Microsoft
HDInsight Connector
Microsoft
OneDrive Connector
Microsoft
SFTP Connector
Microsoft
SAP Connector
Microsoft
SharePoint Online Connector
Microsoft
QuickBooks Connector
Microsoft

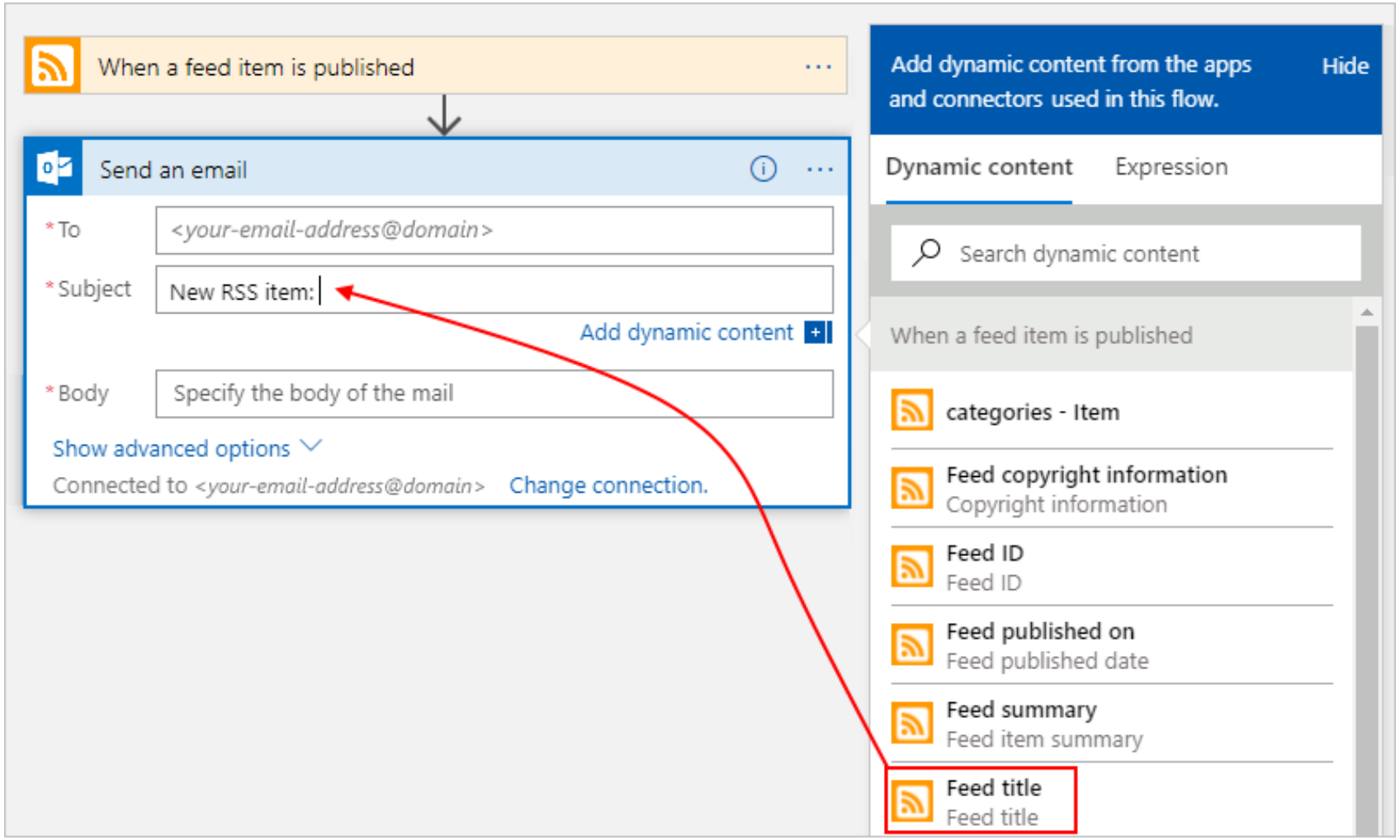
Connector ecosystem



Logic Apps Designer – action configuration

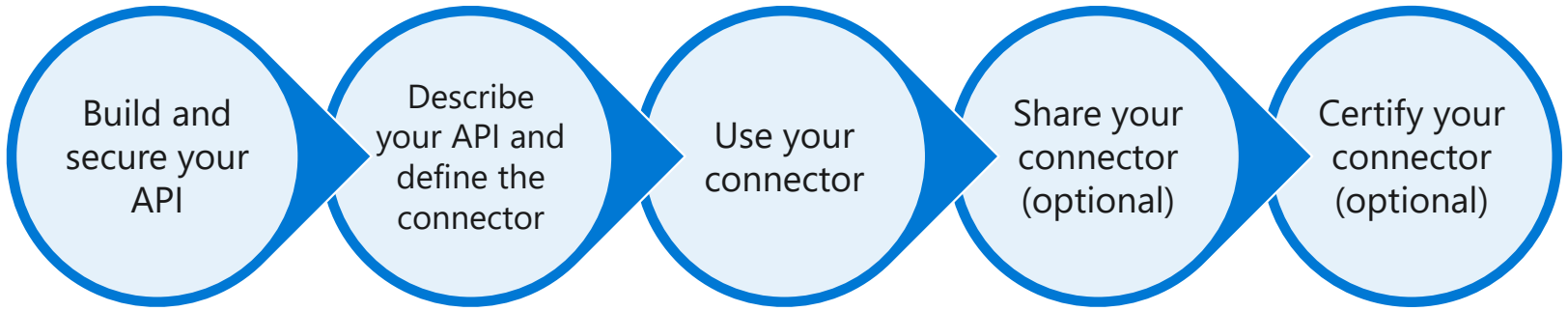


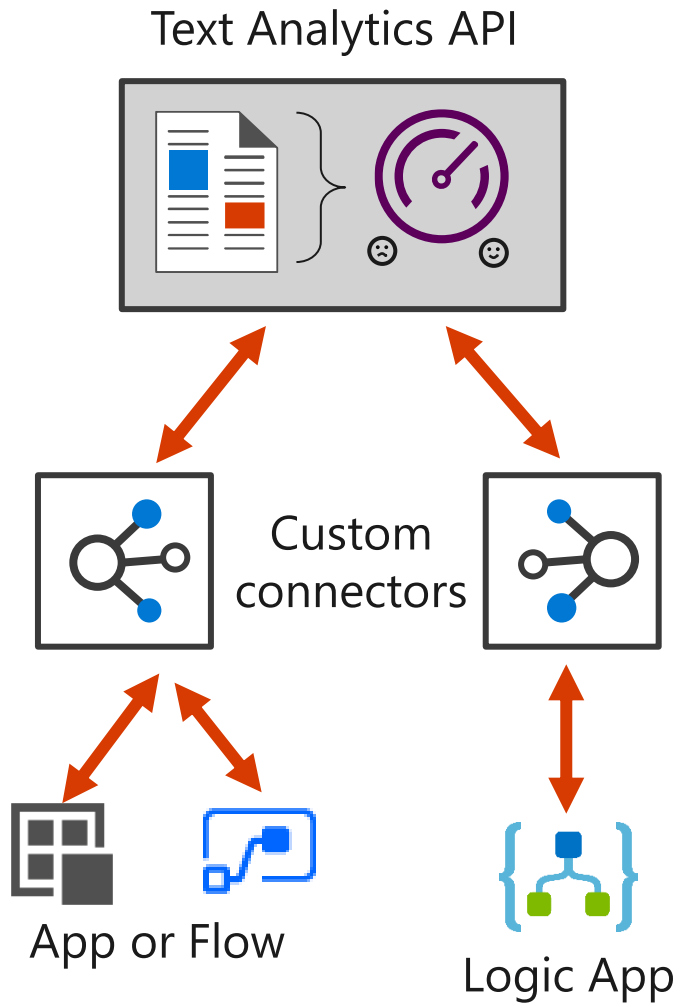
Logic Apps Designer – dynamic content



Custom connectors

- Some APIs, services, and systems are available by using prebuilt connectors
- Build custom connectors:
  - Function-based
  - Custom defined triggers and actions



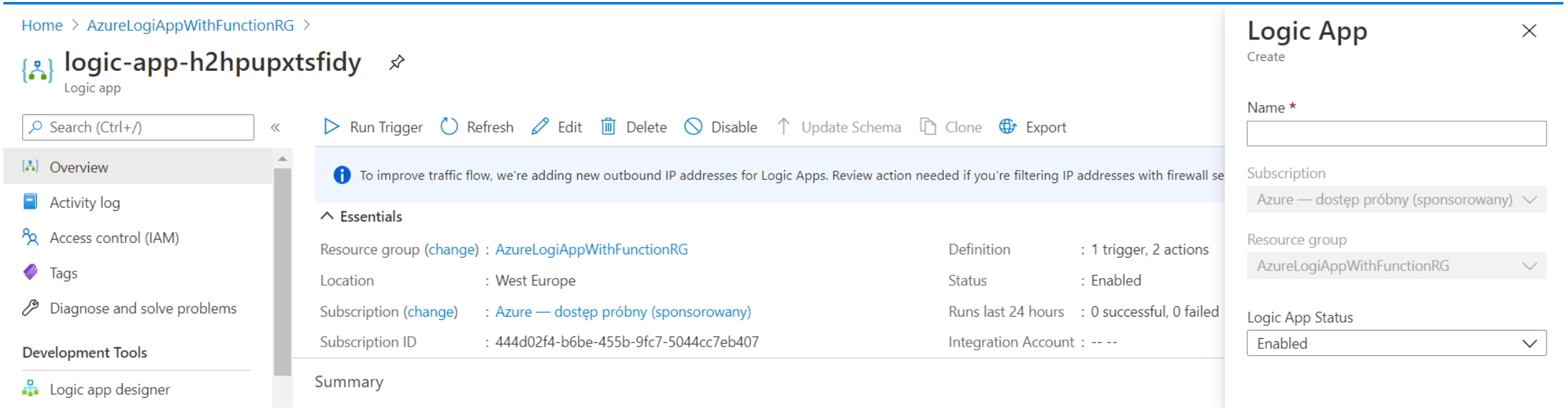


DEMO 2

General Best Practices

- Use descriptive names
- Improves readability
- Transfer knowledge
- You can’t rename Logic Apps … directly
- You can’t rename actions that have dependencies … directly (you can rename it in code)
- If needed, add comments
- Learn from failures

General Best Practices

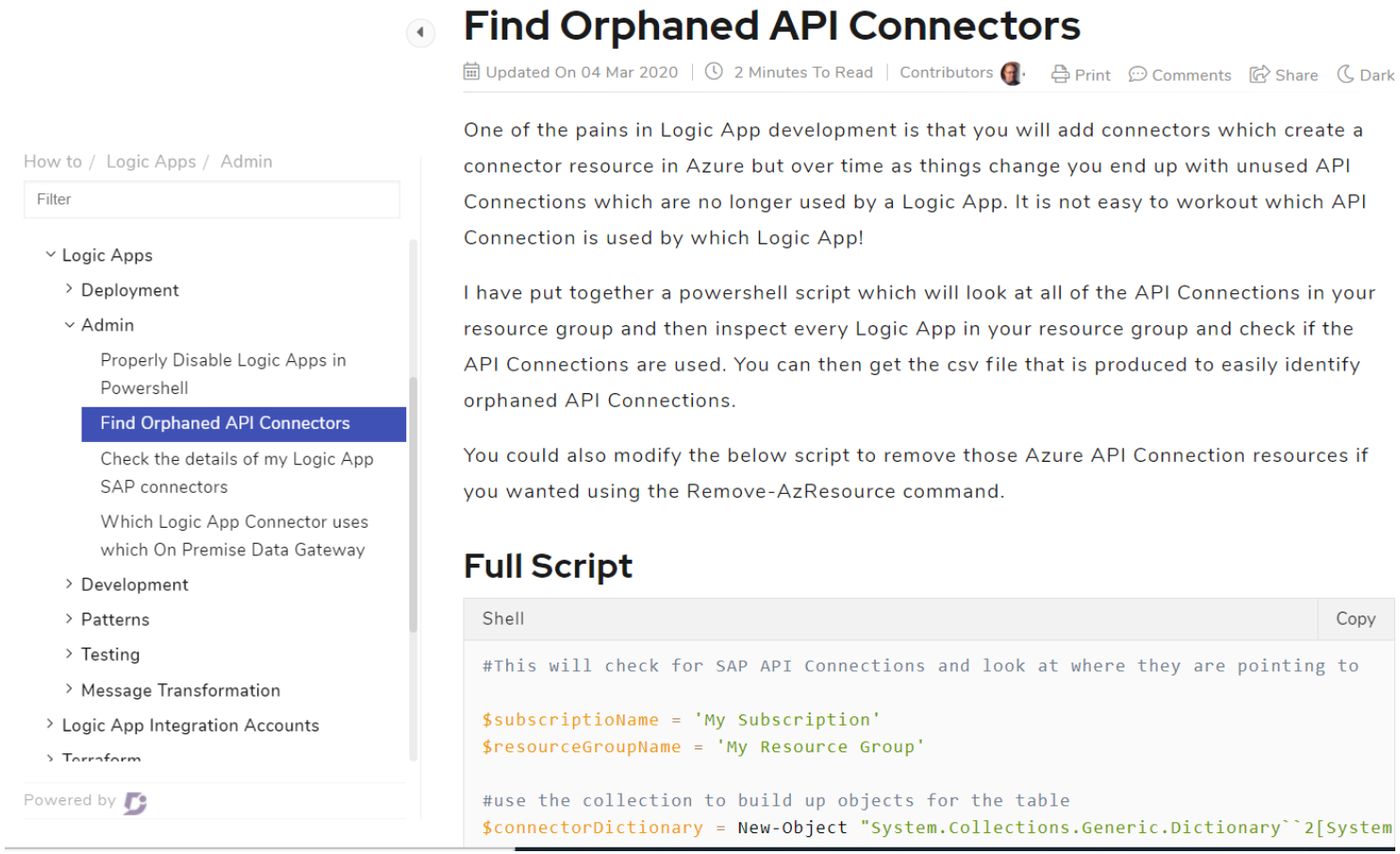


General Best Practices

- Problems with connectors (API connections)
    - Duplicate connections
    - Unused connection
    - Who are using this connection?
- Find orphaned API connectors

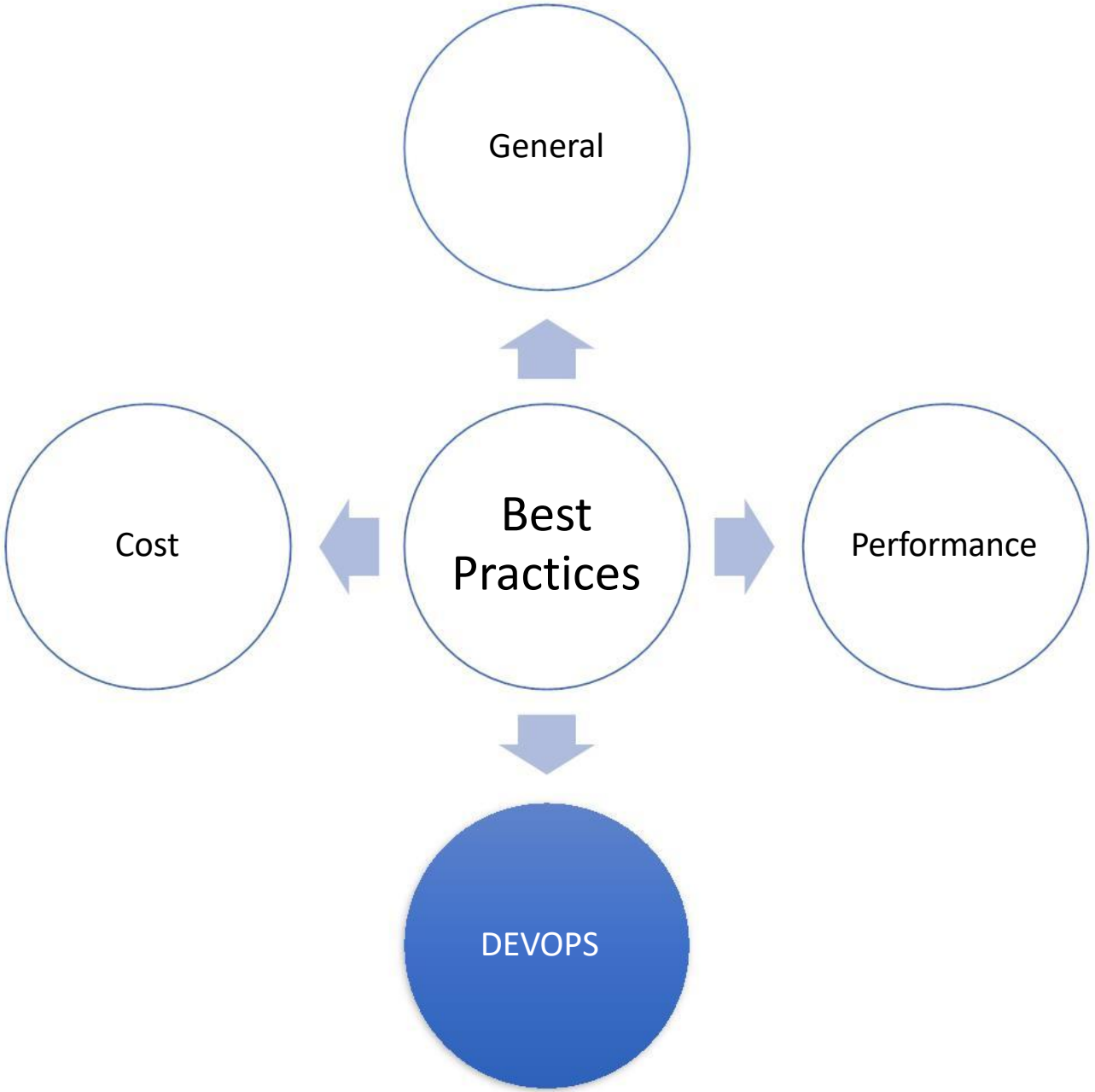
General Best Practices

https://www.integration-playbook.io/docs/find-orphaned-api-connectors



General best practices

- Avoid long-running functions:
  - Functions that run for a long time can time out
- Use queues for cross-function communication:
  - If you require direct communication, consider Durable Functions or Azure Logic Apps
- Write stateless functions:
  - Functions should be stateless
  - State data should be associated with your input and output payloads
- Code defensively:
  - Assume that your function might need to continue from a previous failure point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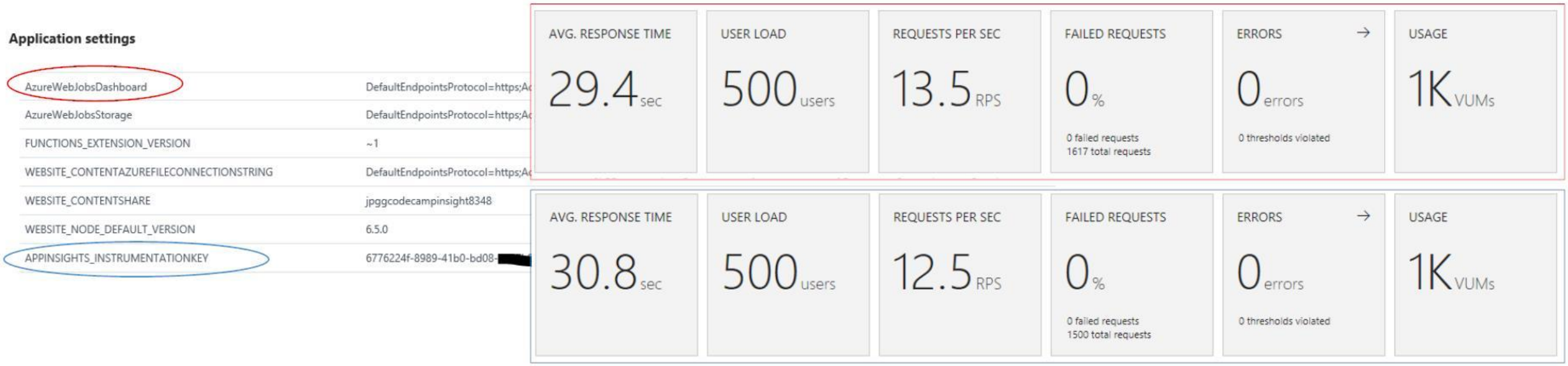
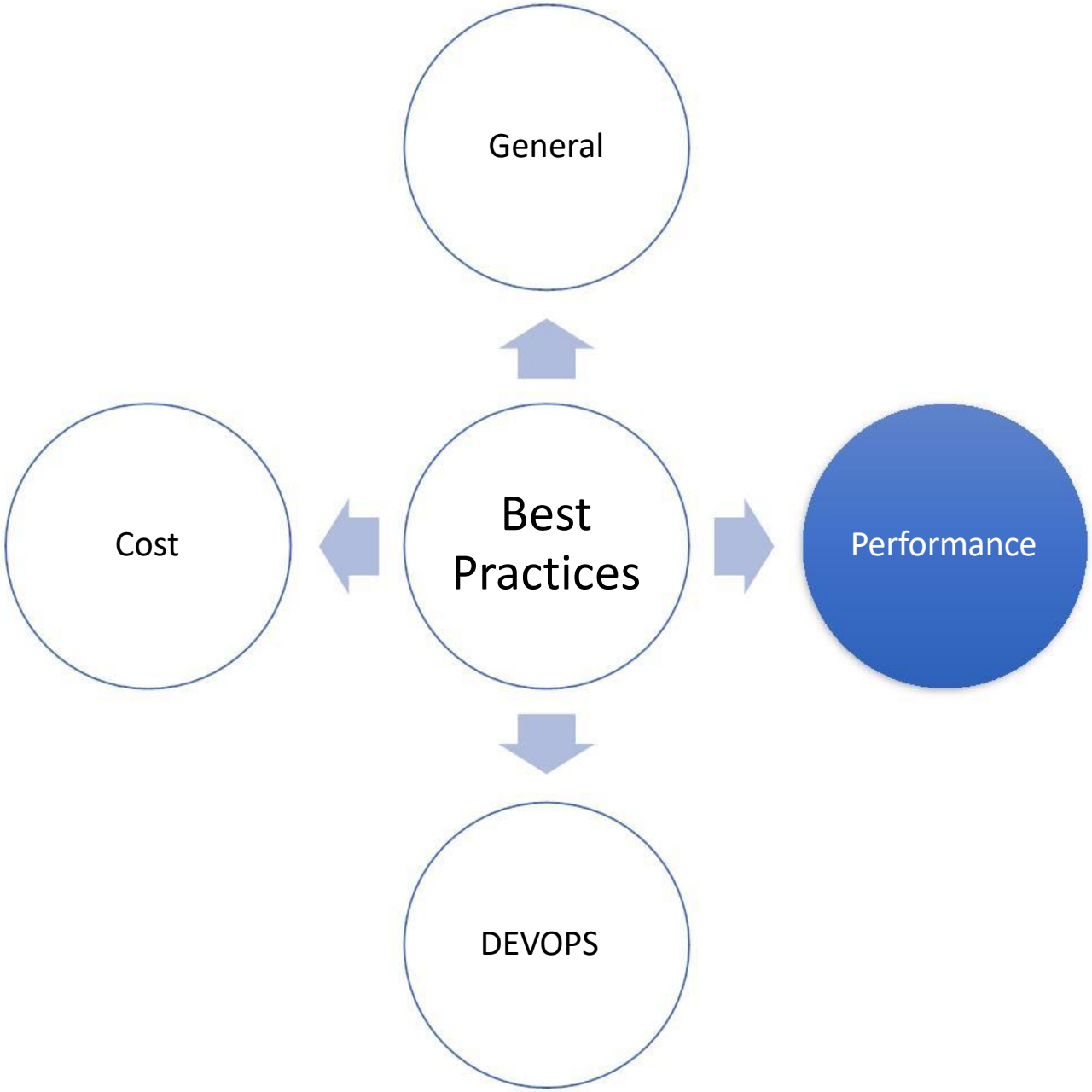
DEV/OPS

- Use source control and CI/CD pipeline
- Always Monitoring!
  - Azure Functions has two built in monitoring solutions
    - WebJobs dashboard & Application Insights



DEVOPS / Application Insights

- **Live Stream**: see a near-live view of what's coming from your Function App
- **Metrics Explorer**: insights on your metrics coming from your Function App
- **Failures**: insights on which things are failing
- **Performance**: information on the count, latency, and more of Function executions
- **Servers**: esource utilization and throughput per server (Putting the servers back in Serverless)
- **Antilytic Portal**: Write custom queries against your data
- **Alerts***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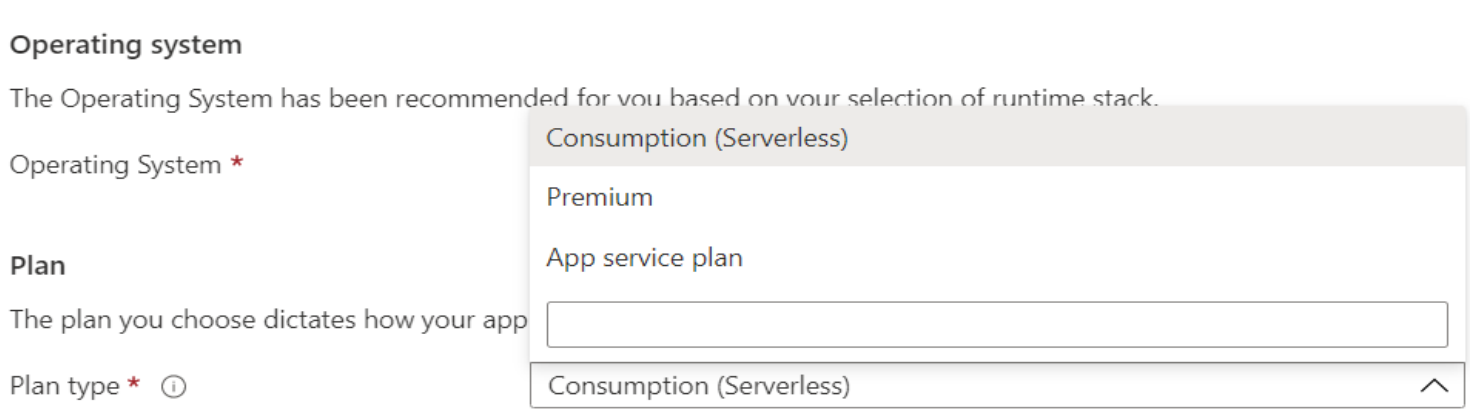
Scale and hosting Function Apps

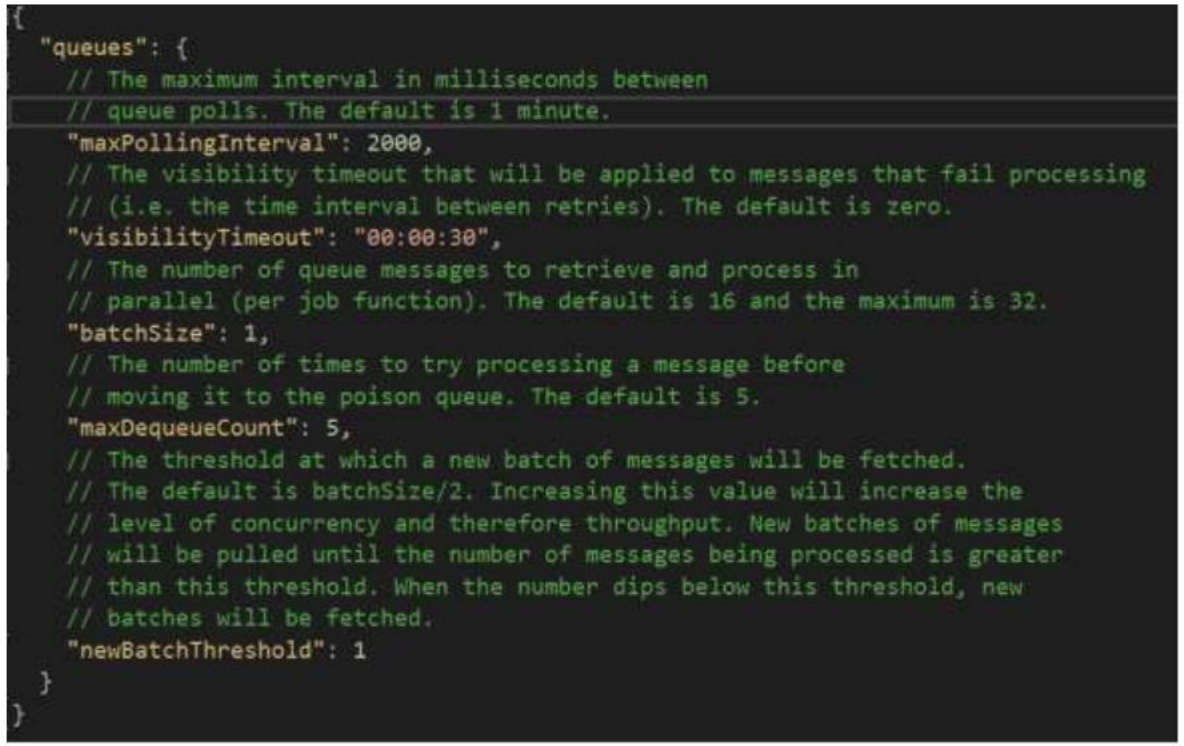
- You can choose between three types of plans:
  - **Consumption**:
    - Instances are dynamically instanced, and you are charged based on compute time
  - **Premium**
    - Instances of the Azure Functions host are added and removed based on the number of incoming events just like the Consumption plan, but provides additional features like: VNet connectivity; unlimited execution duration; and more predictable pricing
  - **App Service plan**:
    - Traditional App Services model used with Web Apps, API Apps, and Mobile Apps
- The type of plan controls:
  - How host instances are scaled out
  - The resources that are

available to each host



Scalability Best Practices

- Receive messages in batch whenever possible/the functions process messages in batches



```
{
  "queues": {
    // The maximum interval in milliseconds between
    // queue polls. The default is 1 minute.
    "maxPollingInterval": 2000,
    // The visibility timeout that will be applied to messages that fail processing
    // (i.e. the time interval between retries). The default is zero.
    "visibilityTimeout": "00:00:30",
    // The number of queue messages to retrieve and process in
    // parallel (per job function). The default is 16 and the maximum is 32.
    "batchSize": 1,
    // The number of times to try processing a message before
    // moving it to the poison queue. The default is 5.
    "maxDequeueCount": 5,
    // The threshold at which a new batch of messages will be fetched.
    // The default is batchSize/2. Increasing this value will increase the
    // level of concurrency and therefore throughput. New batches of messages
    // will be pulled until the number of messages being processed is greater
    // than this threshold. When the number dips below this threshold, new
    // batches will be fetched.
    "newBatchThreshold": 1
  }
}
```

Scalability Best Practices

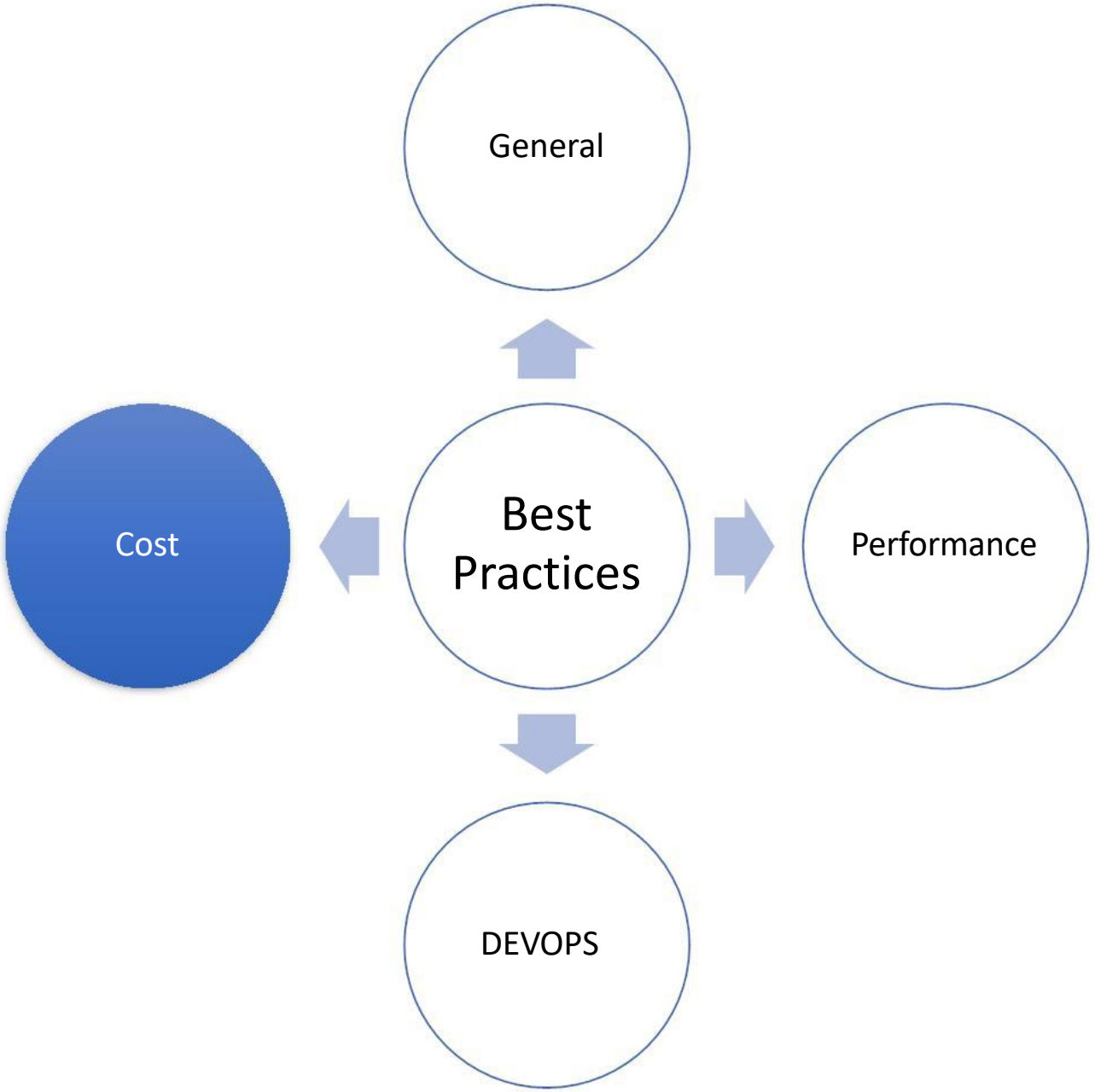
- Use async code but avoid blocking calls
  - C# example: Result V/S await
- Watch your outbound socket connections
  - limit is 300 on Consumption plan + use **connection pooling**

Scalability Best Practices

- Configure host behaviors to better handle concurrency

```
{
    "http": {
        "routePrefix": "api",
        "maxOutstandingRequests": 20,
        "maxConcurrentRequests": 10,
        "dynamicThrottlesEnabled": false
    }
}
```

“scale paradox?“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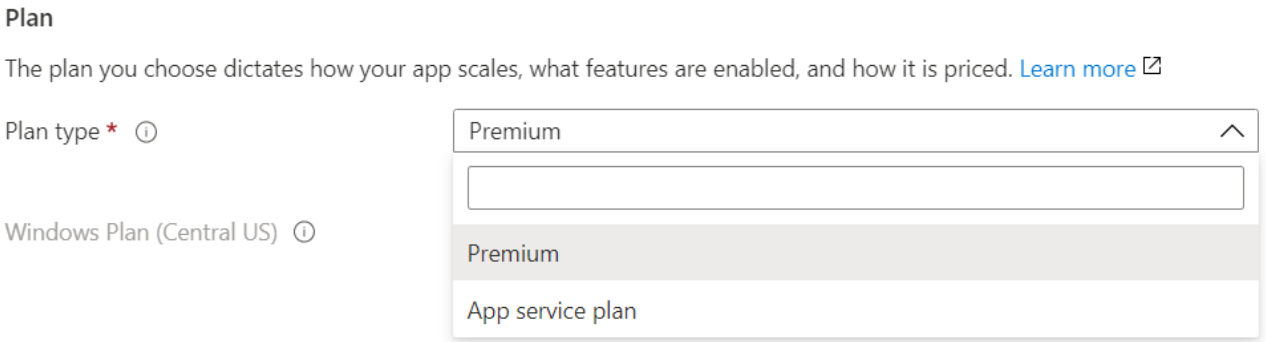
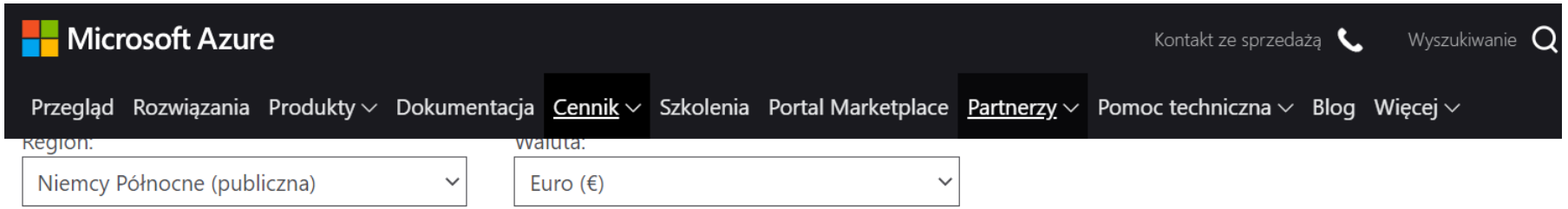
Cost Function Apps

- Consumption plan
  - **Executions**
    - total number of requested executions each month for all functions
    - The first million executions are included **free each month**.
  - **Resource consumption**
    - "**Observed resource consumption**" measured in gigabyte seconds (GB-s).
      - Observed resource consumption is calculated by multiplying average memory size in gigabytes by the time in milliseconds it takes to execute the function.
      - Memory used by a function is measured by rounding up to the nearest 128 MB, up to the maximum memory size of 1,536 MB
      - Execution time calculated by rounding up to the nearest 1 ms.
      - The minimum execution time and memory for a single function execution is 100 ms and 128 mb respectively.
      - Functions pricing includes a **monthly free** grant of 400,000 GB-s.
- App Service plan
  - Same as usual for Web apps

Cost Logic Apps





Szczegóły cennika

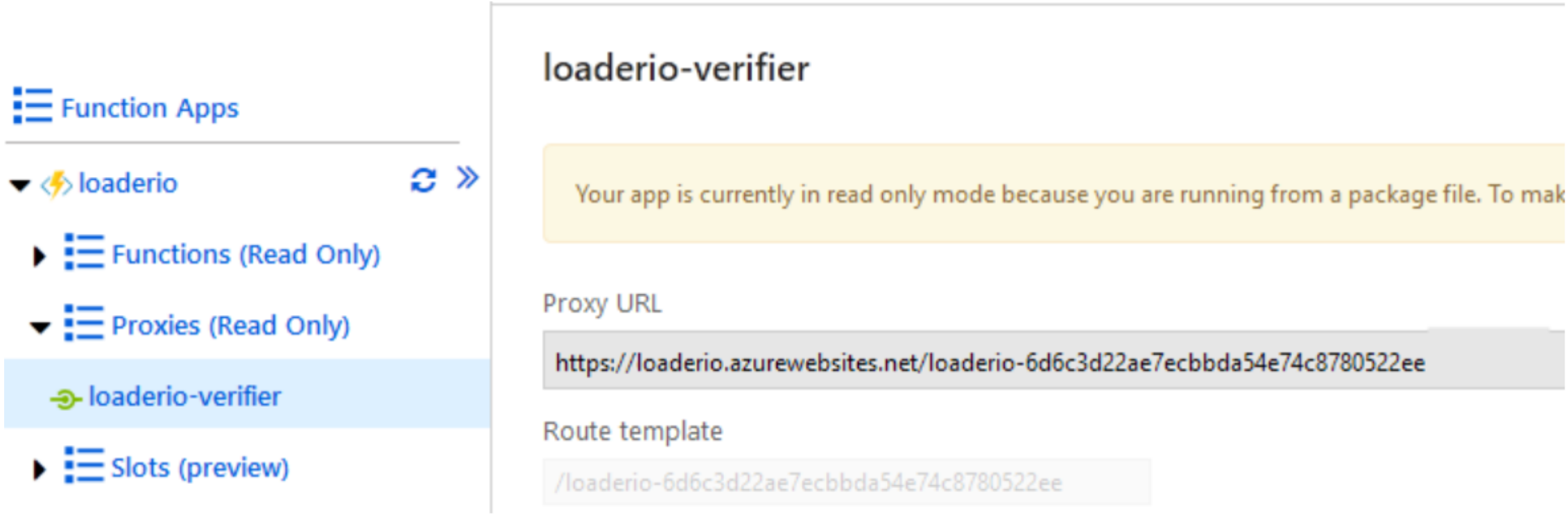
Za każdym razem, gdy definicja aplikacji logiki uruchamia wyzwalacze, wykonania akcji i łączników są mierzone.

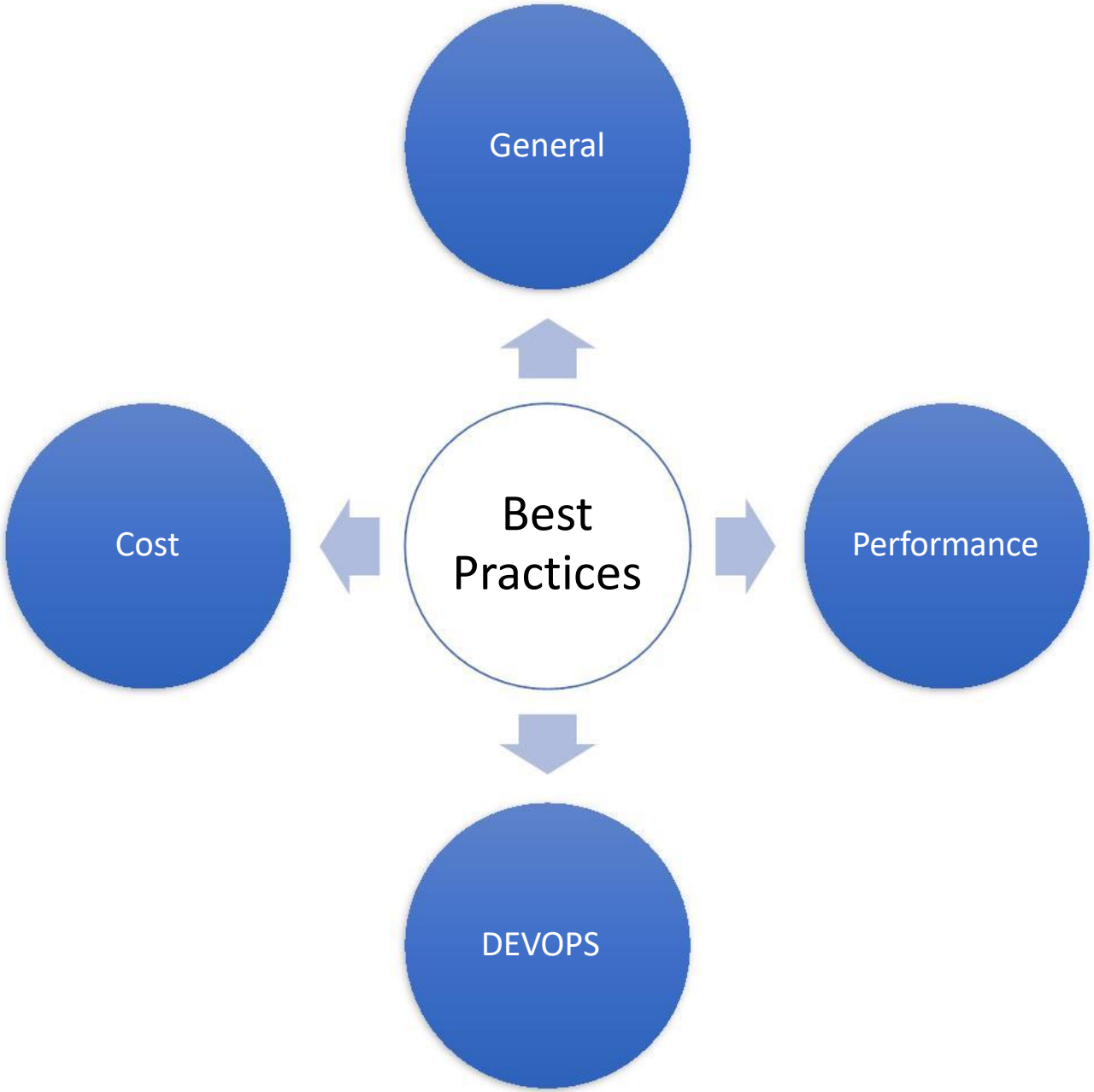
	CENA ZA WYKONANIE
Akcje	€0,000034
Łącznik standardowy	€0,000172
Łącznik przedsiębiorstwa	€0,001375

Loader

https://loader.io/

Loader.io is a FREE load testing service that allows you to stress test your web-apps & api's with thousands of concurrent connections.

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DEMO 3

References

- Optimize the performance and reliability of Azure Functions
  - https://docs.microsoft.com/en-us/azure/azure-functions/functions-best-practices
- https://stackoverflow.com/questions/61101593/azure-logic-app-and-function-app-performance-difference
  - https://stackoverflow.com/questions/61101593/azure-logic-app-and-function-app-performance-difference
- Azure Functions – Significant Improvements in HTTP Trigger Scaling
  - https://www.azurefromthetrenches.com/azure-functions-significant-improvements-in-http-trigger-scaling/
- Azure Function Apps: Performance Considerations
  - https://blogs.msdn.microsoft.com/amitagarwal/2018/04/03/azure-function-apps-performance-considerations/
- Host Health Monitor
  - https://github.com/Azure/azure-functions-host/wiki/Host-Health-Monitor
- Functions pricing
  - https://azure.microsoft.com/en-us/pricing/details/functions/
- Logic apps pricing

  https://azure.microsoft.com/en-us/pricing/details/logic-apps/

References

- Best practices for Azure Functions
https://docs.microsoft.com/en-us/azure/azure-functions/functions-best-practices

- Processing 100,000 events per second on Azure Functions
https://azure.microsoft.com/pl-pl/blog/processing-100-000-events-per-second-on-azure-functions/?ref=msdn

- Serverless samples
https://www.serverlesslibrary.net/sample

- Azure Functions – Prepare for continuous delivery
https://blogs.msdn.microsoft.com/visualstudioalmrangers/2017/09/06/azure-functions-prepare-for-continuous-delivery/

- High throughput Azure Functions on HTTP
https://codez.deedx.cz/projects/high-throughput-azure-functions-on-http/

Q & A

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Thank you for your precious time☺

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